

QistEngine

One utility bill and one mobile-wallet ledger become a defensible installment offer — for the 100 million Pakistanis a credit bureau has never heard of.

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github.com/sohjpeg/qistengine • runs fully offline, two commands

SCORE	300 – 850
■ LOW	720 – 850
■ MEDIUM	640 – 719
■ HIGH	560 – 639
■ VERY HIGH	300 – 559

THE PROBLEM

79% of Pakistani adults have no bank account.

~100M

adults outside the formal
financial system

2.8%

of the world's people

8%

of the world's unbanked adults

A kiryana shopkeeper with twelve years of trading history, a shop the whole neighbourhood knows, and a phone full of daily transactions — still has no credit file. The formal system has no way to see him.

Source: World Bank Global Findex 2021; World Bank / Dawn, "100m Pakistanis without a bank account."

Bureau scoring asks for a loan history they were never allowed to build.

- No first loan → no repayment record → no score → no first loan. The loop never opens.
- Lenders who **do** serve this segment price the uncertainty in: hard collateral, two guarantors, 30–40% effective APR — or they decline.
- The informal alternative — the **committee / BC**, the shopkeeper's ledger book — builds nothing a regulated lender can accept.

US\$57.8B

MSME financing gap — roughly **20% of GDP**

<200k of 3.2M

SMEs with formal credit — though SMEs are 90% of firms and 40% of GDP

THE INSIGHT

The excluded are not invisible. They leave a financial trail — every month.

SIGNAL 1 • THE UTILITY BILL

Nearly every household gets one. Payment punctuality, arrears, reconnection events — a clean 12-month record of **meeting a recurring obligation**.

SIGNAL 2 • THE MOBILE WALLET

80M+ active accounts. JazzCash, EasyPaisa, a hand-kept Digital Khata: inflow, volatility, buffer, merchant/QR mix, **committee (BC) participation**.

The data to underwrite these borrowers already exists. Nobody turns it into a decision an officer can defend. **That is the whole product.**

Source: State Bank of Pakistan branchless banking data; JazzCash / EasyPaisa disclosures, 2024.

An underwriting engine that sits next to a microfinance loan officer.

IN

One electricity bill (PDF or scan) + one wallet statement (CSV or PDF export).

OUT

- A **300–850 score** and risk band
- A **calibrated probability of default** — a frequency, not a rank
- An **exact reason-code ledger** + an adverse-action notice
- A **safe monthly Qist** sized to the cashflow
- A **printable credit memo** for the branch file

APPLICATIONS TODAY

6

APPROVAL RATE

42.9%

MEAN SCORE

598

PORTFOLIO EXPECTED LOSS

Rs 13,231

OVERRIDE RATE

6.7%

governance metric

BAND

All

LOW

MEDIUM

HIGH

VERY HIGH

CITY

All cities

APPLICANT	CITY	SCORE	BAND	REQUESTED	SAFE LIMIT	STATUS	SUBMITTED
AB Ayesha Bibi	Lahore	717	MEDIUM RISK	Rs 60,000	Rs 2,000	Awaiting decision	4 Sept 2026
BA Bilal Ahmed	Karachi	615	HIGH RISK	Rs 120,000	Rs 3,000	Awaiting decision	4 Sept 2026
FK Farhan Khan	Rawalpindi	598	HIGH RISK	Rs 85,000	—	Awaiting decision	4 Sept 2026
NA Nasreen Akhtar	Multan	728	LOW RISK	Rs 45,000	Rs 5,500	Awaiting decision	4 Sept 2026
IA Imran Ali	Faisalabad	319	VERY HIGH RISK	Rs 40,000	—	Awaiting decision	4 Sept 2026
ZG Zubair Gul	Peshawar	662	MEDIUM RISK	Rs 70,000	—	Awaiting decision	4 Sept 2026
TB Tariq Bhatti override	Karachi	533	VERY HIGH RISK	Rs 30,000	—	Approved	3 Sept 2026
TA Tariq Awan	Faisalabad	655	MEDIUM RISK	Rs 60,000	Rs 2,500	Info requested	2 Sept 2026
SC Sadia Chaudhry	Karachi	557	VERY HIGH RISK	Rs 120,000	—	Rejected	2 Sept 2026
KK Kiran Khan	Multan	709	MEDIUM RISK	Rs 30,000	—	Approved	1 Sept 2026
HK Hina Khan	Karachi	556	VERY HIGH RISK	Rs 15,000	—	Rejected	1 Sept 2026

HOW IT WORKS

One pipeline, fully offline, deterministic end to end.

INGEST OCR cascade: PDF text layer → Tesseract → labelled fallback. Statement parser normalises JazzCash / EasyPaisa / khata exports.	FEATURES 26 behavioural features: cashflow level & volatility, buffer, utility discipline, savings, tenure, network.	SCORECARD Calibrated LightGBM + isotonic calibration. SHAP attribution computed per applicant.	SCORE + PD Points-to-double-the-odds scaling. Score from the margin, PD from the calibrated model.	DECISION Qist affordability waterfall, bilingual ledger, adverse-action codes, credit memo.
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- **26 features only** — cashflow, discipline, savings, tenure. No gender, religion, ethnicity, caste, marital status or area proxy ever enters the model.
- **Unreadable input is labelled, never fabricated** — a failed OCR returns a “simulated” extraction at confidence 0.0, visible to the officer.
- **No network on the hot path** — no CDN, no LLM call. Trains and serves on a laptop.

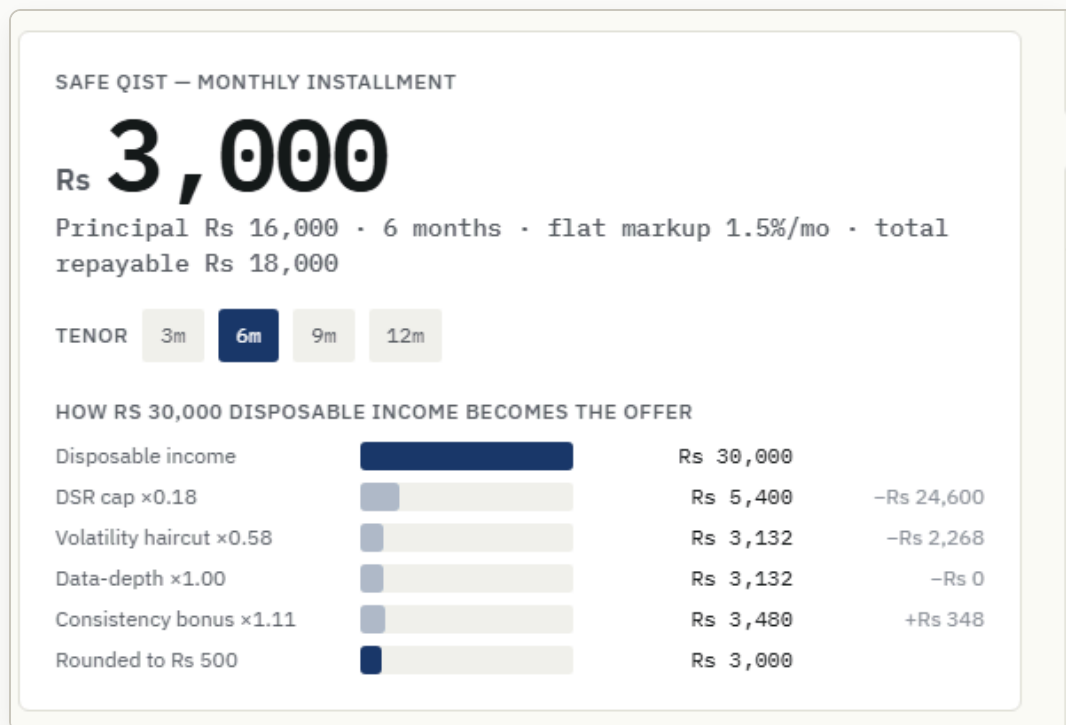
The scorecard reads like a shopkeeper's khata — and it foots exactly.

Qist scorecard ledger		Print credit memo
APP-BILAL · 4 Sept 2026		
DESCRIPTION	CREDIT	DEBIT
Opening balance (population base)	682	
Transfer pattern is unusually concentrated		-48
Spends 76.0% of income, leaving room to repay	+21	
Almost no cash buffer between earnings (0.0%)		-18
Business running for 96 months	+16	
Wallet rarely runs empty	+7	
74.0% of inflow is merchant/QR receipts	+7	
Paid 9 of 12 electricity bills on time	+6	
Income trending up 1.0% over six months	+5	
12 months of billing history on file	+4	
12 smaller factors	+6	
Rounding to nearest point		-1
CLOSING SCORE	615	

High risk · Guarantor or collateral required · PD 8.4%

- Opening balance (population base), then **every factor as a credit or a debit**, closing at the score.
- The score is affine in log-odds; SHAP values are additive in log-odds — so the column **sums to the score exactly**. An officer can add it up by hand.
- Each line is **bilingual** — English and Urdu.
- The largest debits become the **adverse-action codes** on the decline notice.

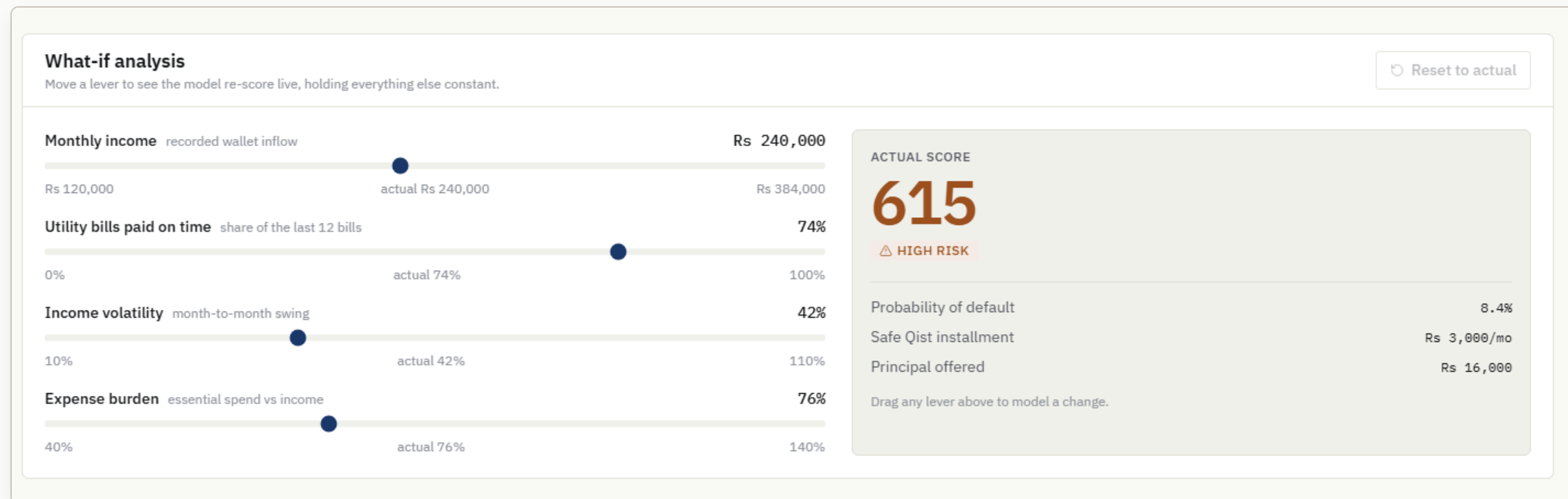
A good score is not permission to lend Rs 50,000.



QistEngine sizes the installment separately from the score:

- **Disposable income**, stepped down by a debt-service cap, a **volatility haircut**, a data-depth factor, and a consistency bonus.
- Floored to Rs 500; ineligible below Rs 2,000 — the engine says **“not yet,” not a bad number.**
- Thin-file applicants still get an offer, with the confidence discount shown on its face.

What-if analysis — the officer moves a lever, the model re-scores live.



“Pay two more of your next bills on time and you move from HIGH to MEDIUM.”
The applicant leaves with a concrete path, not just a rejection — and the officer never touches a spreadsheet. Nothing is written back to the application.

Protected attributes are forbidden as features — the engine still audits itself.

- Gender, religion, ethnicity, caste, marital status, area — never features. They exist in the synthetic data only so the audit can run.
- Every run emits a **four-fifths (80%) rule** check. Gender and city tier pass. Two **livelihood** groups are flagged — genuine cashflow risk, not a protected class — with documented mitigations.

Model card

qistengine-scorecard-v1.0.0

ROC-AUC	0.819	PR-AUC	0.460	KS statistic	0.488
Gini	0.638	Brier score	0.099	Logistic baseline AUC	0.817
Base default rate	14.5%	Features	26	Trained	3 Sept 2026

FAIRNESS AUDIT

FOUR-FIFTHS RULE: FLAGGED

Approve = score ≥ 560. Portfolio approval 71.5%, false-positive rate 22.1%. Gender and city tier pass; the flags below are livelihood groups where genuine cashflow risk lowers approval.

FLAGGED GROUP	APPROVAL	RATIO VS BEST	OBSERVED DEFAULT
archetype · daily wage worker	33%	0.35	34%
archetype · ride hailing driver	73%	0.79	12%

Calibrated LightGBM · isotonic calibration · 26 features

ROC-AUC	0.819
KS statistic	0.488
Gini	0.638
Brier score (calibration)	0.099
Logistic baseline AUC	0.817
Base default rate	14.5%

- **Calibrated**, not just ranked — a predicted 8% PD means 8 in 100, which is what the affordability maths needs.
- Beats the transparent logistic baseline and, more importantly, separates the **tails** far better.
- Trained on a synthetic portfolio from a **documented data-generating process**: archetype cashflow distributions, non-linear repayment dynamics, measurement noise.

BUILT TO BE TRUSTED

Every claim on the screen can be checked.

- **“Not a regulated credit decision”** on every decision surface.
- **Adverse-action codes** for every decline — ECOA-style.
- **Deterministic** — seed 42, byte-identical training metrics on a re-run.
- **Fully offline** — no CDN, no LLM, no telemetry.
- **34 passing tests**; every dependency pinned to an exact version.
- **Two commands** from a clean clone to a running app.

```
git clone https://github.com/sohjpeg/qistengine.git && cd qistengine && bash run-dev.sh
```

This is a prototype. The model learns from synthetic data — because no lender hands a student team its repayment book.

REAL & PORTABLE

The feature engineering, the scorecard maths, the calibration, the fairness audit, the ingestion pipeline, the officer UI — all of it moves to production unchanged.

WHAT A PILOT REPLACES

Exactly one thing: the synthetic repayment outcomes, swapped for a partner MFI's actual history. Then recalibrate. The architecture doesn't move.

We would rather show a defensible pipeline on honest synthetic data than a magic number on data we can't explain.

From prototype to a merchant's first loan.

- **1** Partner with one microfinance institution or BNPL lender.
- **2** Score their last 12 months of disbursements against known outcomes → recalibrate on real defaults.
- **3** Shadow mode for a quarter: the engine recommends, the officer decides, we compare.
- **4** Measure: approval rate at constant loss, time-to-decision, officer override rate.

The goal: a merchant's first Rs 15,000 loan, decided in minutes, and defensible on paper to a regulator.

Score the data they already generate.

QistEngine — alternative credit scoring for Pakistan's unbanked merchants and micro-entrepreneurs.

Sohaib Amir Bukhari · Asma Imran

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